

Notes on Multinomial Model

Max Leung

April 14, 2026

1 General Multinomial Model

$$y_i \in \{1, \dots, m\}$$

Define m dummy variables

$$y_{ij} = \begin{cases} 1 & y_i = j \\ 0 & y_i \neq j \end{cases} \quad \text{for } j \in \{1, \dots, m\}$$

Model p_{ij}

$$p_{ij} := Pr(y_i = j | \mathbf{x}_i) = F_j(\mathbf{x}_i, \boldsymbol{\beta}) \quad \text{for } j \in \{1, \dots, m\}$$

where $\sum_{j=1}^m p_{ij} = 1$

1.1 ML Estimation

Log likelihood function

$$f(y_i) = p_{i1}^{y_{i1}} \cdots p_{im}^{y_{im}} = \prod_{j=1}^m p_{ij}^{y_{ij}}$$

$$\begin{aligned} \ln[L_N(\boldsymbol{\beta})] &= \ln\left[\prod_{i=1}^N f(y_i)\right] && \text{assume independence} \\ &= \sum_{i=1}^N \ln[f(y_i)] \\ &= \sum_{i=1}^N \ln\left[\prod_{j=1}^m F_j(\mathbf{x}_i, \boldsymbol{\beta})^{y_{ij}}\right] \\ &= \sum_{i=1}^N \sum_{j=1}^m y_{ij} \ln[F_j(\mathbf{x}_i, \boldsymbol{\beta})] \end{aligned}$$

$$\begin{aligned}
\frac{\partial \ln[L_N(\boldsymbol{\beta})]}{\partial \boldsymbol{\beta}} &= \frac{\partial \sum_{i=1}^N \sum_{j=1}^m y_{ij} \ln[F_j(\mathbf{x}_i, \boldsymbol{\beta})]}{\partial \boldsymbol{\beta}} \\
&= \sum_{i=1}^N \sum_{j=1}^m y_{ij} \frac{\partial \ln[F_j(\mathbf{x}_i, \boldsymbol{\beta})]}{\partial \boldsymbol{\beta}} \\
&= \sum_{i=1}^N \sum_{j=1}^m y_{ij} \frac{1}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}}
\end{aligned}$$

FOC

$$\sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \Big|_{\hat{\boldsymbol{\beta}}} = \mathbf{0}$$

1.2 Consistency of MLE

It requires the correct specification of $F_j(\mathbf{x}_i, \boldsymbol{\beta})$

1.3 Asymptotic Distribution of MLE

$$\hat{\boldsymbol{\beta}} \sim_a N(\boldsymbol{\beta}_0, [\mathbf{I}(\boldsymbol{\beta}_0)]^{-1}) = N(\boldsymbol{\beta}_0, [\sum_{i=1}^N \sum_{j=1}^m \{F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)^{-1} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} - \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0}\}]^{-1})$$

$$\begin{aligned}
\mathbf{I}(\boldsymbol{\beta}_0) &:= -\mathbb{E}\left[\frac{\partial^2 \ln L_N(\boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \Big| \mathbf{x}_i\right] = \mathbb{E}\left[\frac{\partial \ln L_N(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \cdot \frac{\partial \ln L_N(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \Big| \mathbf{x}_i\right] \\
&= -\mathbb{E}\left[\frac{\partial \sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}}}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \Big| \mathbf{x}_i\right] \\
&= -\sum_{i=1}^N \sum_{j=1}^m \mathbb{E}\left[\frac{\partial \frac{y_{ij}}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}}}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \Big| \mathbf{x}_i\right] \\
&= -\sum_{i=1}^N \sum_{j=1}^m \mathbb{E}\left[y_{ij} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})^{-1}}{\partial \boldsymbol{\beta}'} + \frac{y_{ij}}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big| \mathbf{x}_i\right]_{\boldsymbol{\beta}_0} \\
&= -\sum_{i=1}^N \sum_{j=1}^m \mathbb{E}\left[-y_{ij} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} F_j(\mathbf{x}_i, \boldsymbol{\beta})^{-2} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} + \frac{y_{ij}}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big| \mathbf{x}_i\right]_{\boldsymbol{\beta}_0} \\
&= -\sum_{i=1}^N \sum_{j=1}^m \left\{ -\mathbb{E}[y_{ij} | \mathbf{x}_i] \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} F_j(\mathbf{x}_i, \boldsymbol{\beta})^{-2} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} + \frac{\mathbb{E}[y_{ij} | \mathbf{x}_i]}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \right\} \Big|_{\boldsymbol{\beta}_0} \\
&= -\sum_{i=1}^N \sum_{j=1}^m \left\{ -F_j(\mathbf{x}_i, \boldsymbol{\beta}_0) \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} F_j(\mathbf{x}_i, \boldsymbol{\beta})^{-2} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} + \frac{F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)}{F_j(\mathbf{x}_i, \boldsymbol{\beta})} \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \right\} \Big|_{\boldsymbol{\beta}_0} \\
&= -\sum_{i=1}^N \sum_{j=1}^m \left\{ -F_j(\mathbf{x}_i, \boldsymbol{\beta}_0) \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \Big|_{\boldsymbol{\beta}_0} F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)^{-2} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} + \frac{F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)}{F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)} \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \right\} \\
&= -\sum_{i=1}^N \sum_{j=1}^m \left\{ -F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)^{-1} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} + \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \right\} \\
&= \sum_{i=1}^N \sum_{j=1}^m \left\{ F_j(\mathbf{x}_i, \boldsymbol{\beta}_0)^{-1} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \frac{\partial F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} - \frac{\partial^2 F_j(\mathbf{x}_i, \boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \Big|_{\boldsymbol{\beta}_0} \right\}
\end{aligned}$$

1.4 McFadden's Pseudo- R^2

$$\begin{aligned}
R_{multi}^2 &= 1 - \frac{\ln L_{fit}}{\ln L_0} \\
&= \frac{\ln L_0 - \ln L_{fit}}{\ln L_0} \\
&= \frac{\ln L_0 - \ln L_{fit}}{\ln L_0 - 0} \\
&= \frac{\ln L_0 - \ln L_{fit}}{\ln L_0 - \ln L_{max}} \\
&= \frac{\ln L_{fit} - \ln L_0}{\ln L_{max} - \ln L_0}
\end{aligned}$$

$$L_{max} = \prod_{i=1}^N \prod_{j=1}^m p_{ij}^{y_{ij}} = \prod_{i=1}^N 0^0 \dots 1^1 \dots 0^0 = \prod_{i=1}^N 1 \dots 1 \dots 1 = \prod_{i=1}^N 1 = 1$$

1.5 Special Case: Conditional Logit Model

if $F_j(\mathbf{x}_i, \boldsymbol{\beta}) = F_{softmax}(\mathbf{x}'_{ij}\boldsymbol{\beta})$

$$p_{ij} := Pr(y_i = j | \mathbf{x}_i) = F_{softmax}(\mathbf{x}'_{ij}\boldsymbol{\beta}) = \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta})} \quad \text{for } j \in \{1, \dots, m\}$$

1.5.1 Gradient Vector

$$\begin{aligned}
\frac{\partial p_{ij}}{\partial \boldsymbol{\beta}} &= \frac{\partial \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta})}}{\partial \boldsymbol{\beta}} \\
&= \frac{\partial \exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \left(\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \right)^{-1} + \frac{\partial (\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}))^{-1}}{\partial \boldsymbol{\beta}} \exp(\mathbf{x}'_{ij}\boldsymbol{\beta}) \\
&= \exp(\mathbf{x}'_{ij}\boldsymbol{\beta}) \mathbf{x}_{ij} \left(\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \right)^{-1} - \left(\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \right)^{-2} \left[\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \mathbf{x}_{il} \right] \exp(\mathbf{x}'_{ij}\boldsymbol{\beta}) \\
&= \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta})} \mathbf{x}_{ij} - \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{(\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}))^2} \left[\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta}) \mathbf{x}_{il} \right] \\
&= \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta})} \mathbf{x}_{ij} - \frac{\exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\sum_{l=1}^m \exp(\mathbf{x}'_{il}\boldsymbol{\beta})} \left[\sum_{l=1}^m \frac{\exp(\mathbf{x}'_{il}\boldsymbol{\beta})}{\sum_{k=1}^m \exp(\mathbf{x}'_{ik}\boldsymbol{\beta})} \mathbf{x}_{il} \right] \\
&= p_{ij} \mathbf{x}_{ij} - p_{ij} \left[\sum_{l=1}^m p_{il} \mathbf{x}_{il} \right] \\
&= p_{ij} \mathbf{x}_{ij} - p_{ij} \bar{\mathbf{x}}_i \\
&= p_{ij} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)
\end{aligned}$$

$\bar{\mathbf{x}}_i = \sum_{l=1}^m p_{il} \mathbf{x}_{il}$

$$\frac{\partial \ln[L_N(\boldsymbol{\beta})]}{\partial \boldsymbol{\beta}} = \sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{p_{ij}} \frac{\partial p_{ij}}{\partial \boldsymbol{\beta}} = \sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{p_{ij}} p_{ij} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i) = \sum_{i=1}^N \sum_{j=1}^m y_{ij} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)$$

1.5.2 Asymptotic Distribution

$$\begin{aligned}
\frac{\partial^2 \ln[L_N(\boldsymbol{\beta})]}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} &= \frac{\sum_{i=1}^N \sum_{j=1}^m \partial y_{ij}(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)}{\partial \boldsymbol{\beta}'} = \sum_{i=1}^N \sum_{j=1}^m \frac{\partial y_{ij}(\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)}{\partial \boldsymbol{\beta}'} \\
&= - \sum_{i=1}^N \sum_{j=1}^m \frac{\partial y_{ij} \bar{\mathbf{x}}_i}{\partial \boldsymbol{\beta}'} \\
&= - \sum_{i=1}^N \sum_{j=1}^m \frac{\partial y_{ij} \sum_{l=1}^m p_{il} \mathbf{x}_{il}}{\partial \boldsymbol{\beta}'} \\
&= - \sum_{i=1}^N \sum_{j=1}^m y_{ij} \sum_{l=1}^m \frac{\partial p_{il}}{\partial \boldsymbol{\beta}'} \mathbf{x}_{il} \\
&= - \sum_{i=1}^N \sum_{j=1}^m y_{ij} \sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i)' \mathbf{x}_{il} \\
&= - \sum_{i=1}^N \left[\sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i) \mathbf{x}_{il}' \right] \sum_{j=1}^m y_{ij} \\
&= - \sum_{i=1}^N \sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i) \mathbf{x}_{il}' \quad \sum_j y_{ij} = 1 \\
&= - \sum_{i=1}^N \sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i) (\mathbf{x}_{il} - \bar{\mathbf{x}}_i)'
\end{aligned}$$

The last line is due to $\sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i) \mathbf{x}_{il}' = \sum_{l=1}^m (p_{il} \mathbf{x}_{il} - p_{il} \bar{\mathbf{x}}_i) \mathbf{x}_{il}' = (\sum_{l=1}^m p_{il} \mathbf{x}_{il} - [\sum_{l=1}^m p_{il}] \bar{\mathbf{x}}_i) \mathbf{x}_{il}' = (\bar{\mathbf{x}}_i - 1 \cdot \bar{\mathbf{x}}_i) \mathbf{x}_{il}' = \mathbf{0}$

Thus, the asymptotic variance is $[\mathbf{I}(\boldsymbol{\beta}_0)]^{-1} := [-\mathbb{E}(\frac{\partial^2 \ln[L_N(\boldsymbol{\beta})]}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} | \mathbf{x}_i)]^{-1} = [\sum_{i=1}^N \sum_{l=1}^m p_{il} (\mathbf{x}_{il} - \bar{\mathbf{x}}_i) (\mathbf{x}_{il} - \bar{\mathbf{x}}_i)']^{-1}$

1.6 Special Case: Multinomial Logit Model

if $F_j(\mathbf{x}_i, \boldsymbol{\beta}) = F_{\text{softmax}}(\mathbf{x}_i' \boldsymbol{\beta}_j)$

$$p_{ij} := \Pr(y_i = j | \mathbf{x}_i) = F_{\text{softmax}}(\mathbf{x}_i' \boldsymbol{\beta}_j) = \frac{\exp(\mathbf{x}_i' \boldsymbol{\beta}_j)}{\sum_{l=1}^m \exp(\mathbf{x}_i' \boldsymbol{\beta}_l)} \quad \text{for } j \in \{1, \dots, m\}$$

1.6.1 Gradient Vector

$$\begin{aligned}
\frac{\partial p_{ij}}{\partial \beta_j} &= \frac{\partial \frac{\exp(\mathbf{x}'_i \beta_j)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)}}{\partial \beta_j} \\
&= \frac{\partial \exp(\mathbf{x}'_i \beta_j)}{\partial \beta_j} \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-1} + \frac{\partial (\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l))^{-1}}{\partial \beta_j} \exp(\mathbf{x}'_i \beta_j) \\
&= \exp(\mathbf{x}'_i \beta_j) \mathbf{x}_i \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-1} - \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-2} \left[\sum_{l=1}^m \frac{\partial \exp(\mathbf{x}'_i \beta_l)}{\partial \beta_j} \right] \exp(\mathbf{x}'_i \beta_j) \\
&= \exp(\mathbf{x}'_i \beta_j) \mathbf{x}_i \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-1} - \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-2} \exp(\mathbf{x}'_i \beta_j) \mathbf{x}_i \exp(\mathbf{x}'_i \beta_j) \\
&= \frac{\exp(\mathbf{x}'_i \beta_j)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)} \mathbf{x}_i - \left(\frac{\exp(\mathbf{x}'_i \beta_j)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)} \right)^2 \mathbf{x}_i \\
&= p_{ij} \mathbf{x}_i - p_{ij}^2 \mathbf{x}_i
\end{aligned}$$

$$\begin{aligned}
\frac{\partial p_{ij}}{\partial \beta_k} &= \frac{\partial \frac{\exp(\mathbf{x}'_i \beta_j)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)}}{\partial \beta_k} \\
&= \exp(\mathbf{x}'_i \beta_j) \frac{\partial (\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l))^{-1}}{\partial \beta_k} \\
&= -\exp(\mathbf{x}'_i \beta_j) \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-2} \sum_{l=1}^m \frac{\partial \exp(\mathbf{x}'_i \beta_l)}{\partial \beta_k} \\
&= -\exp(\mathbf{x}'_i \beta_j) \left(\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l) \right)^{-2} \exp(\mathbf{x}'_i \beta_k) \mathbf{x}_i \\
&= -\frac{\exp(\mathbf{x}'_i \beta_j)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)} \frac{\exp(\mathbf{x}'_i \beta_k)}{\sum_{l=1}^m \exp(\mathbf{x}'_i \beta_l)} \mathbf{x}_i \\
&= -p_{ij} p_{ik} \mathbf{x}_i
\end{aligned}$$

Thus,

$$\frac{\partial p_{ij}}{\partial \beta_l} = 1(l = j)p_{ij}\mathbf{x}_i - p_{ij}p_{il}\mathbf{x}_i = p_{ij}(1(l = j) - p_{il})\mathbf{x}_i$$

$$\begin{aligned} \frac{\partial \ln[L_N(\boldsymbol{\beta})]}{\partial \beta_l} &= \sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{p_{ij}} \frac{\partial p_{ij}}{\partial \beta_l} \\ &= \sum_{i=1}^N \sum_{j=1}^m \frac{y_{ij}}{p_{ij}} p_{ij}(1(l = j) - p_{il})\mathbf{x}_i \\ &= \sum_{i=1}^N \mathbf{x}_i \sum_{j=1}^m y_{ij}(1(l = j) - p_{il}) \\ &= \sum_{i=1}^N \mathbf{x}_i [-p_{il} \sum_{j \neq l}^m y_{ij} + y_{il}(1 - p_{il})] \\ &= \sum_{i=1}^N \mathbf{x}_i [-p_{il} \sum_{j \neq l}^m y_{ij} + y_{il} - y_{il}p_{il}] \\ &= \sum_{i=1}^N \mathbf{x}_i [-p_{il} \sum_{j=1}^m y_{ij} + y_{il}] \\ &= \sum_{i=1}^N \mathbf{x}_i [y_{il} - p_{il}] \end{aligned} \quad \text{As } \sum_{j=1}^m y_{ij} = 1$$

1.6.2 Asymptotic Distribution

$$\begin{aligned} \frac{\partial^2 \ln[L_N(\boldsymbol{\beta})]}{\partial \beta_j \partial \beta'_k} &= \frac{\partial \sum_{i=1}^N \mathbf{x}_i [y_{ij} - p_{ij}]}{\partial \beta'_k} \\ &= - \sum_{i=1}^N \frac{\partial p_{ij}}{\partial \beta'_k} \mathbf{x}_i \\ &= - \sum_{i=1}^N p_{ij}(1(k = j) - p_{ik})\mathbf{x}_i \mathbf{x}'_i \end{aligned}$$

Thus, the asymptotic variance is $[\mathbf{I}(\boldsymbol{\beta}_0)]^{-1} := [-\mathbb{E}(\frac{\partial^2 \ln[L_N(\boldsymbol{\beta})]}{\partial \beta_j \partial \beta'_k} | \mathbf{x}_i)]^{-1} = [\sum_{i=1}^N p_{ij}(1(k = j) - p_{ik})\mathbf{x}_i \mathbf{x}'_i]^{-1}$

1.7 Special Case: Universal Logit Model

if $F_j(\mathbf{x}_i, \boldsymbol{\beta}) = F_{softmax}(\ln[V_{ij}])$

$$p_{ij} := \Pr(y_i = j | \mathbf{x}_i) = F_{softmax}(\ln[V_{ij}]) = \frac{\exp(\ln[V_{ij}])}{\sum_{l=1}^m \exp(\ln[V_{il}])} = \frac{V_{ij}}{\sum_{l=1}^m V_{il}} \quad \text{for } j \in \{1, \dots, m\}$$

2 Additive Random Utility Model

Suppress i ,

$$U_j = V_j + \varepsilon_j \quad j \in \{1, \dots, m\}$$

$$\begin{aligned} p_j &:= Pr(y = j | \mathbf{x}) \\ &= Pr(U_j > U_k \ \forall k \neq j | \mathbf{x}) \\ &= Pr(V_j + \varepsilon_j > V_k + \varepsilon_k \ \forall k \neq j | \mathbf{x}) \\ &= Pr(V_j - V_k > \varepsilon_k - \varepsilon_j \ \forall k \neq j | \mathbf{x}) \\ &= Pr(\varepsilon_k - \varepsilon_j < V_j - V_k \ \forall k \neq j | \mathbf{x}) \\ &= Pr(\varepsilon_k < \varepsilon_j + V_j - V_k \ \forall k \neq j | \mathbf{x}) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\varepsilon_j + V_j - V_1} \dots \int_{-\infty}^{\varepsilon_j + V_j - V_{j-1}} \int_{-\infty}^{\varepsilon_j + V_j - V_{j+1}} \dots \int_{-\infty}^{\varepsilon_j + V_j - V_m} f(\varepsilon_1, \dots, \varepsilon_m) \partial \varepsilon_m \dots \partial \varepsilon_{j+1} \partial \varepsilon_{j-1} \dots \partial \varepsilon_1 \partial \varepsilon_j \end{aligned}$$

2.1 Special Case: Conditional Logit Model, Multinomial Logit Model

If $\varepsilon_k \ \forall k$ are independent and all follow Type 1 Extreme Value Distribution (or log Weibull Distribution). It can be shown $p_j := Pr(y = j | \mathbf{x}) = \frac{\exp(V_j)}{\sum_{i=1}^m \exp(V_i)}$ by direct integration method. If $V_{ij} = \mathbf{x}'_{ij} \boldsymbol{\beta}$, it is Conditional Logit Model. If $V_{ij} = \mathbf{x}'_i \boldsymbol{\beta}_j$, it is Multinomial Logit Model.

2.2 Special Case: Generalized Extreme Value Model (McFadden, 1978)

Nested Logit Model is a special case of GEV model. Conditional Logit and Multinomial Logit Models are special case of Nested Logit Model.

If $\boldsymbol{\varepsilon}$ follows (multivariate) GEV distribution i.e., $F(\boldsymbol{\varepsilon}) = F(\varepsilon_1, \dots, \varepsilon_m) = \exp[-G(e^{-\varepsilon_1}, \dots, e^{-\varepsilon_m})]$ where $G(\cdot)$ is a non-negative homogeneous function of degree 1, $\rightarrow \infty$ when any of its arguments $\rightarrow \infty$ and the n th partial derivatives of G are nonnegative for odd n and nonpositive for even n , then $p_j := Pr(y = j | \mathbf{x}) = e^{V_j} \frac{G_j(e^{V_1}, \dots, e^{V_m})}{G(e^{V_1}, \dots, e^{V_m})}$ where G_j is derivative of the j argument. Such a formula is a consequence of the following theorem.

Theorem (McFadden)

If $\boldsymbol{\varepsilon}$ follows (multivariate) GEV distribution as defined above. Define $Z := \max_j [V_j + \varepsilon_j]$. Then Z follows $(\ln G(e^{V_1}, \dots, e^{V_m}), 1)$ -Gumble distribution.

Proof: Let F_Z be the cdf of Z ,

$$\begin{aligned} F_Z(z) &= Pr(Z \leq z) = Pr(\max_j [V_j + \varepsilon_j] \leq z) \\ &= Pr(V_j + \varepsilon_j \leq z \ \forall j) = Pr(\varepsilon_j \leq z - V_j \ \forall j) \\ &= \exp[-G(e^{-(z-V_1)}, \dots, e^{-(z-V_m)})] \\ &= \exp[-e^{-z} G(e^{V_1}, \dots, e^{V_m})] \quad \text{As } G \text{ is homogeneous function of degree 1} \\ &= \exp[-e^{-z} e^{\ln G(e^{V_1}, \dots, e^{V_m})}] \\ &= \exp[-e^{-(z - \ln G(e^{V_1}, \dots, e^{V_m}))}] \end{aligned}$$

The last line is the cdf of $(\ln G(e^{V_1}, \dots, e^{V_m}), 1)$ -Gumble random variable. Note that the cdf of $(0, 1)$ -Gumble (Type 1 Extreme Value or log Weibull) random variable is $\exp[-e^{-z}]$. Q.E.D.

By the property of $(\ln G(e^{V_1}, \dots, e^{V_m}), 1)$ -Gumble random variable, $\mathbb{E}[\max_j[V_j + \varepsilon_j]] = \ln G(e^{V_1}, \dots, e^{V_m}) + \gamma$ where γ is the Euler's constant (around 0.5772). Therefore,

$$\frac{\partial \mathbb{E}[\max_i[V_i + \varepsilon_i]]}{\partial V_j} = \frac{\partial \ln G(e^{V_1}, \dots, e^{V_m}) + \gamma}{\partial V_j} = \frac{e^{V_j} G_j(e^{V_1}, \dots, e^{V_m})}{G(e^{V_1}, \dots, e^{V_m})}$$

2.2.1 Nested Logit Model

The following derivation is based on the specialization of the general framework of GEV model. For the simple problem like car-red-blue-bus problem, we can derive $p_{k|j}$ by direct integration (see Amemiya, 1985 page 301-302).

$$F(\varepsilon_{11}, \dots, \varepsilon_{1K_1}; \dots; \varepsilon_{J1}, \dots, \varepsilon_{JK_J}) = \exp[-G(e^{-\varepsilon_{11}}, \dots, e^{-\varepsilon_{1K_1}}; \dots; e^{-\varepsilon_{J1}}, \dots, e^{-\varepsilon_{JK_J}})]$$

$$p_{jk} := \Pr(y_{jk} = 1 | \mathbf{x}) = e^{V_{jk}} \frac{G_{jk}(e^{V_{11}}, \dots, e^{V_{1K_1}}; \dots; e^{V_{J1}}, \dots, e^{V_{JK_J}})}{G(e^{V_{11}}, \dots, e^{V_{1K_1}}; \dots; e^{V_{J1}}, \dots, e^{V_{JK_J}})}$$

For Nested Logit Model, we specify $G(\cdot)$ as

$$G(\mathbf{Y}) = G(Y_{11}, \dots, Y_{1K_1}; \dots; Y_{J1}, \dots, Y_{JK_J}) = \sum_{j=1}^J a_j \left(\sum_{k=1}^{K_j} Y_{jk}^{1/\rho_j} \right)^{\rho_j}$$

where Y_{jk} can be $e^{V_{jk}}$ or $e^{-\varepsilon_{jk}}$. If $J = 1$ and $K_1 = 2$ and $a_1 = 1$, $G(\mathbf{Y}) = (Y_1^{1/\rho} + Y_2^{1/\rho})^\rho$. The resulting $F(\varepsilon_1, \varepsilon_2) = \exp[-(e^{-\varepsilon_1/\rho} + e^{-\varepsilon_2/\rho})^\rho]$ is the cdf of Gumbel's Type B bivariate extreme-value random variable. It can be shown that $\text{Corr}(\varepsilon_1, \varepsilon_2) = 1 - \rho^2$. If $\rho = 1$, $F(\varepsilon_1, \varepsilon_2) = \exp[-e^{-\varepsilon_1}] \exp[-e^{-\varepsilon_2}]$, which is a product of the cdfs of Type 1 Extreme Value random variables. Nested Logit Model thus reduces to Conditional Logit or Multinomial Logit Model. Note also that it is the case of binary Logit Model. $\Pr(y = 1 | \mathbf{x}) = e^{V_1} \frac{1}{e^{V_1} + e^{V_2}} = \frac{1}{1 + e^{V_2 - V_1}} = \frac{1}{1 + e^{-\mathbf{x}'_i \boldsymbol{\theta}_0}}$.

If $J = 1$ and $K_1 = 3$ and $a_1 = 1$, $G(\mathbf{Y}) = (Y_1^{1/\rho} + Y_2^{1/\rho} + Y_3^{1/\rho})^\rho$. $\Pr(y = 1 | \mathbf{x}) = e^{V_1} \frac{(e^{V_1/\rho} + e^{V_2/\rho} + e^{V_3/\rho})^{\rho-1} e^{(1-\rho)V_1/\rho}}{(e^{V_1/\rho} + e^{V_2/\rho} + e^{V_3/\rho})^\rho} = \frac{e^{V_1/\rho}}{e^{V_1/\rho} + e^{V_2/\rho} + e^{V_3/\rho}}$. Therefore, $\frac{\Pr(y=1|\mathbf{x})}{\Pr(y=2|\mathbf{x})} = \frac{\exp(V_1/\rho)}{\exp(V_2/\rho)} = \exp((V_1 - V_2)/\rho)$.

For ρ_j in $\sum_{j=1}^J a_j (\sum_{k=1}^{K_j} Y_{jk}^{1/\rho_j})^{\rho_j}$, it is not exactly equal to the correlation coefficients. It can be regarded as an approximation though.

$$\begin{aligned}
G_{j'k'} &= \frac{\partial G(\mathbf{Y})}{\partial Y_{j'k'}} \\
&= \frac{\partial \sum_{j=1}^J a_j (\sum_{k=1}^{K_j} Y_{jk}^{1/\rho_j})^{\rho_j}}{\partial Y_{j'k'}} \\
&= a_{j'} \frac{\partial (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}}}{\partial Y_{j'k'}} \\
&= a_{j'} \rho_{j'} (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}-1} \sum_{k=1}^{K_{j'}} \frac{\partial Y_{j'k}^{1/\rho_{j'}}}{\partial Y_{j'k'}} \\
&= a_{j'} \rho_{j'} (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}-1} \frac{\partial Y_{j'k'}^{1/\rho_{j'}}}{\partial Y_{j'k'}} \\
&= a_{j'} \rho_{j'} (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}-1} (1/\rho_{j'}) Y_{j'k'}^{1/\rho_{j'}-1} \\
&= a_{j'} (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}-1} Y_{j'k'}^{1/\rho_{j'}-1}
\end{aligned}$$

$$Y_{j'k'} G_{j'k'} = a_{j'} (\sum_{k=1}^{K_{j'}} Y_{j'k}^{1/\rho_{j'}})^{\rho_{j'}-1} Y_{j'k'}^{1/\rho_{j'}}$$

$$\begin{aligned}
p_{jk} &= Y_{jk} \frac{G_{jk}}{G(\mathbf{Y})} \\
&= \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j-1} Y_{jk}^{1/\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}}
\end{aligned}$$

$$\begin{aligned}
p_j &:= \sum_{k=1}^{K_j} p_{jk} = \sum_{k=1}^{K_j} \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j-1} Y_{jk}^{1/\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}} \\
&= \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j-1} \sum_{k=1}^{K_j} Y_{jk}^{1/\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}} \\
&= \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}}
\end{aligned}$$

$$\begin{aligned}
p_{k|j} &= \frac{p_{jk}}{p_j} = \frac{\frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j-1} Y_{jk}^{1/\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}}}{\frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}}} \\
&= \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j-1} Y_{jk}^{1/\rho_j}}{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j}} \\
&= \frac{Y_{jk}^{1/\rho_j}}{\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j}}
\end{aligned}$$

Substitute back

$$\begin{aligned}
(e^{V_{jl}})^{1/\rho_j} &= (e^{\mathbf{z}'_j \boldsymbol{\alpha} + \mathbf{x}'_{jl} \boldsymbol{\beta}_j})^{1/\rho_j} \\
&= e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j + \mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j} \\
&= e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}
\end{aligned}$$

$$\begin{aligned}
\sum_{l=1}^{K_j} (e^{V_{jl}})^{1/\rho_j} &= \sum_{l=1}^{K_j} e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j} \\
&= e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} \sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j} \\
&= e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} \exp(\ln[\sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}]) \\
&= e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} \exp(l_j)
\end{aligned}$$

$$\begin{aligned}
(\sum_{l=1}^{K_j} (e^{V_{jl}})^{1/\rho_j})^{\rho_j} &= (\exp(\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j) \exp(l_j))^{\rho_j} \\
&= (\exp(\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j + l_j))^{\rho_j} \\
&= \exp(\mathbf{z}'_j \boldsymbol{\alpha} + \rho_j l_j)
\end{aligned}$$

$$\begin{aligned}
p_j &= \frac{a_j (\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j})^{\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} Y_{ml}^{1/\rho_m})^{\rho_m}} \\
&= \frac{a_j (\sum_{l=1}^{K_j} e^{V_{jl}^{1/\rho_j}})^{\rho_j}}{\sum_{m=1}^J a_m (\sum_{l=1}^{K_m} e^{V_{ml}^{1/\rho_m}})^{\rho_m}} \\
&= \frac{a_j \exp(\mathbf{z}'_j \boldsymbol{\alpha} + \rho_j l_j)}{\sum_{m=1}^J a_m \exp(\mathbf{z}'_m \boldsymbol{\alpha} + \rho_m l_m)}
\end{aligned}$$

$$\begin{aligned}
p_{k|j} &= \frac{Y_{jk}^{1/\rho_j}}{\sum_{l=1}^{K_j} Y_{jl}^{1/\rho_j}} \\
&= \frac{e^{V_{jk}^{1/\rho_j}}}{\sum_{l=1}^{K_j} e^{V_{jl}^{1/\rho_j}}} \\
&= \frac{e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} e^{\mathbf{x}'_{jk} \boldsymbol{\beta}_j / \rho_j}}{\sum_{l=1}^{K_j} e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}} \\
&= \frac{e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} e^{\mathbf{x}'_{jk} \boldsymbol{\beta}_j / \rho_j}}{e^{\mathbf{z}'_j \boldsymbol{\alpha} / \rho_j} \sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}} \\
&= \frac{e^{\mathbf{x}'_{jk} \boldsymbol{\beta}_j / \rho_j}}{\sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}}
\end{aligned}$$

Thus,

$$\begin{aligned}
p_{jk} &= p_{k|j} p_j \\
&= \frac{e^{\mathbf{x}'_{jk} \boldsymbol{\beta}_j / \rho_j}}{\sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl} \boldsymbol{\beta}_j / \rho_j}} \frac{a_j \exp(\mathbf{z}'_j \boldsymbol{\alpha} + \rho_j l_j)}{\sum_{m=1}^J a_m \exp(\mathbf{z}'_m \boldsymbol{\alpha} + \rho_m l_m)}
\end{aligned}$$

where $l_j = \ln[\sum_{l=1}^{K_j} e^{\mathbf{x}'_{jl}\boldsymbol{\beta}_j/\rho_j}]$ is log sum

2.2.2 FIML Estimation of Nested Logit Model

$$\begin{aligned}
f(\mathbf{y}_i) &= \prod_{j=1}^J \prod_{k=1}^{K_j} p_{ijk}^{y_{ijk}} \\
&= \prod_{j=1}^J \prod_{k=1}^{K_j} (p_{ik|j} p_{ij})^{y_{ijk}} \\
&= \prod_{j=1}^J \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}} p_{ij}^{y_{ijk}} \\
&= \prod_{j=1}^J \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}} \prod_{k=1}^{K_j} p_{ij}^{y_{ijk}} \\
&= \prod_{j=1}^J \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}} \cdot p_{ij}^{\sum_{k=1}^{K_j} y_{ijk}} \\
&= \prod_{j=1}^J p_{ij}^{\sum_{k=1}^{K_j} y_{ijk}} \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}} \\
&= \prod_{j=1}^J p_{ij}^{y_{ij}} \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}}
\end{aligned}$$

$$\begin{aligned}
\ln[L(\boldsymbol{\alpha}, \boldsymbol{\beta}_j, \rho_j)] &= \ln\left[\prod_{i=1}^N f(\mathbf{y}_i)\right] && \text{assume independence of } i \\
&= \ln\left[\prod_{i=1}^N \prod_{j=1}^J \prod_{k=1}^{K_j} p_{ik|j}^{y_{ijk}} p_{ij}^{y_{ijk}}\right] \\
&= \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} \ln[p_{ik|j}^{y_{ijk}} p_{ij}^{y_{ijk}}] \\
&= \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} \{y_{ijk} \ln[p_{ik|j}] + y_{ijk} \ln[p_{ij}]\} \\
&= \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} y_{ijk} \ln[p_{ik|j}] + \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} y_{ijk} \ln[p_{ij}] \\
&= \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} y_{ijk} \ln[p_{ik|j}] + \sum_{i=1}^N \sum_{j=1}^J \ln[p_{ij}] \underbrace{\sum_{k=1}^{K_j} y_{ijk}}_{y_{ij}} \\
&= \sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} y_{ijk} \ln\left[\frac{e^{\mathbf{x}'_{ijk}\boldsymbol{\beta}_j/\rho_j}}{\sum_{l=1}^{K_j} e^{\mathbf{x}'_{ijl}\boldsymbol{\beta}_j/\rho_j}}\right] + \sum_{i=1}^N \sum_{j=1}^J y_{ij} \ln\left[\frac{\exp(\mathbf{z}'_{ij}\boldsymbol{\alpha} + \rho_j l_{ij})}{\sum_{m=1}^J \exp(\mathbf{z}'_{im}\boldsymbol{\alpha} + \rho_m l_{im})}\right]
\end{aligned}$$

2.2.3 LIML Estimation of Nested Logit Model

It is less efficient compared to FIML

First Stage

Estimate $\boldsymbol{\beta}_j/\rho_j$ with maximizing log likelihood $\sum_{i=1}^N \sum_{j=1}^J \sum_{k=1}^{K_j} y_{ijk} \ln\left[\frac{e^{\mathbf{x}'_{ijk}\boldsymbol{\beta}_j/\rho_j}}{\sum_{l=1}^{K_j} e^{\mathbf{x}'_{ijl}\boldsymbol{\beta}_j/\rho_j}}\right]$

Second Stage

Estimate α and ρ_j with maximizing log likelihood $\sum_{i=1}^N \sum_{j=1}^J y_{ij} \ln \left[\frac{\exp(\mathbf{z}'_{ij} \alpha + \rho_j l_{ij})}{\sum_{m=1}^J \exp(\mathbf{z}'_{im} \alpha + \rho_m l_{im})} \right]$

Get β_j by $\beta_j / \rho_j \cdot \rho_j$

2.3 Random Parameters Logit Model

If $V_{ij} = \mathbf{x}'_{ij} \beta_i$ i.e., Conditional Logit Model with heterogeneous beta.

$$U_{ij} = \mathbf{x}'_{ij} \beta_i + \varepsilon_{ij}$$

where

$$\beta_i \sim N(\beta, \Gamma_\beta)$$

It can be written as

$$\begin{aligned} U_{ij} &= \mathbf{x}'_{ij} (\beta + \mathbf{u}_i) + \varepsilon_{ij} & \mathbf{u}_i &\sim N(\mathbf{0}, \Gamma_\beta) \\ &= \mathbf{x}'_{ij} \beta + \underbrace{(\mathbf{x}'_{ij} \mathbf{u}_i + \varepsilon_{ij})}_{v_{ij}} \end{aligned}$$

$Cov(v_{ij}, v_{ik}) = Cov(\mathbf{x}'_{ij} \mathbf{u}_i + \varepsilon_{ij}, \mathbf{x}'_{ik} \mathbf{u}_i + \varepsilon_{ik}) = Cov(\mathbf{x}'_{ij} \mathbf{u}_i, \mathbf{x}'_{ik} \mathbf{u}_i) = \mathbf{x}'_{ij} \Gamma_\beta \mathbf{x}_{ik}$ for $j \neq k$ i.e., Conditional Logit Model with correlated error v_{ij} across alternative.

2.3.1 Maximum Simulated Estimation

$$\begin{aligned} p_{ij} &:= Pr(y_i = j | \mathbf{x}_i) = \int Pr(y_i = j, \beta_i | \mathbf{x}_i, \beta, \Gamma_\beta) d\beta_i \\ &= \int Pr(y_i = j | \beta_i, \mathbf{x}_i, \beta, \Gamma_\beta) \phi(\beta_i | \mathbf{x}_i, \beta, \Gamma_\beta) d\beta_i \\ &= \int Pr(y_i = j | \beta_i, \mathbf{x}_i) \phi(\beta_i | \beta, \Gamma_\beta) d\beta_i \\ &= \int \frac{\exp(\mathbf{x}'_{ij} \beta_i)}{\sum_{l=1}^m \exp(\mathbf{x}'_{il} \beta_i)} \phi(\beta_i | \beta, \Gamma_\beta) d\beta_i \\ &= \mathbb{E} \left[\frac{\exp(\mathbf{x}'_{ij} \beta_i)}{\sum_{l=1}^m \exp(\mathbf{x}'_{il} \beta_i)} \right] \end{aligned}$$

Simulate p_{ij} by drawing β_i from $\phi(\beta_i | \beta, \Gamma_\beta)$ S times

$$\hat{p}_{ij} = S^{-1} \sum_{s=1}^S \frac{\exp(\mathbf{x}'_{ij} \beta_i^s)}{\sum_{l=1}^m \exp(\mathbf{x}'_{il} \beta_i^s)}$$

Thus, the log simulated likelihood function is

$$\begin{aligned} \ln[\hat{L}] &= \ln \left[\prod_{i=1}^N \prod_{j=1}^m \hat{p}_{ij}^{y_{ij}} \right] \\ &= \sum_{i=1}^N \sum_{j=1}^m y_{ij} \ln[\hat{p}_{ij}] \\ &= \sum_{i=1}^N \sum_{j=1}^m y_{ij} \ln \left[S^{-1} \sum_{s=1}^S \frac{\exp(\mathbf{x}'_{ij} \beta_i^s)}{\sum_{l=1}^m \exp(\mathbf{x}'_{il} \beta_i^s)} \right] \end{aligned}$$

3 References

Cameron, A. C., & Trivedi, P. K. (2005). *Microeconometrics: Methods and Applications*.